Asymptotic Inversion of the Gamma and Barnes G-Functions

Lambert-Core Transseries and a General Reversion Calculus

Vladimir Reshetnikov
ProveIt research note

3 September 2026

Abstract

For a large positive value y, this article derives explicit all-orders asymptotic transseries for the large real solutions of $\Gamma(x) = y$ and $G(x) = y$, where G is the Barnes G-function. The leading inversions are not ordinary power-series reversions. After taking logarithms, their dominant phases are respectively $x(\log x - 1)$ and $\frac{1}{2}x^2(\log x - \frac{3}{2})$, and their exact inverse cores are expressed by the Lambert W-function. The lower-order Stirling terms then generate a second, exponentially finer grid of correction blocks.

For the upper inverse Gamma branch, put

$$R = \log\left(\frac{y}{\sqrt{2\pi}}\right), \qquad w = \mathrm{W}_0(R/\mathrm{e}), \qquad T = \frac{R}{w}, \qquad \Lambda = \log T = w + 1.$$

We prove

$$\Gamma_+^{-1}(y) \sim \frac{1}{2} + T + \sum_{n\geq 1} \frac{d_{2n-1}(\Lambda)}{T^{2n-1}},$$

where $d_{2n-1} \in \Lambda^{-1}\mathbb{Q}[\Lambda^{-1}]$, give a triangular coefficient recurrence, prove remainder transfer from Stirling's series, and list the first five nonzero blocks through T^{-9}. The first two blocks agree with the finite expansion previously proved by Gerhold, Hubalek, and Tomovski; the recurrence continues it to arbitrary order. We also record the ordinary convergent inverse series for the second positive branch tending to zero.

For Barnes G, it is cleaner to solve $G(z + 1) = y$. With

$$C_G = \zeta'(-1), \quad R = \log y - C_G, \quad v = \mathrm{W}_0(4R/\mathrm{e}^3), \quad U = 2\sqrt{R/v}, \quad \lambda = \frac{v+1}{2} = \log U - 1,$$

we obtain

$$z \sim U + \sum_{n\geq 0} \frac{b_n(\lambda)}{U^n}, \qquad G_+^{-1}(y) = 1 + z.$$

The coefficient functions are explicit rational Laurent polynomials in λ with coefficients in $\mathbb{Q}[\frac{1}{2}\log(2\pi)]$. A logarithmic term in the forward Barnes expansion is resonant with the core slope; it forces the nonzero coefficientwise limit

$$\sum_{n\geq 1} e_n(\lambda) q^n \longrightarrow \sqrt{1 + q^2/6} - 1 \qquad (\lambda \to \infty),$$

which explains the limiting constants $1/12, -1/288, 1/3456, \ldots$ in the odd inverse-power blocks.

The two examples are instances of a general theory for phases $ax^p(\log x - \beta) + \Psi(x)$. We develop exact Lambert-core inversion, a Hahn-series implicit-function theorem, a support-semigroup calculus, triangular and Newton–Hensel algorithms, a Lagrange–Bürmann reduction to near-identity reversion, denominator bounds, resonant logarithmic sectors, complex branch labels, residual error certificates, and transport of exponentially small sectors. Formal, Poincaré-asymptotic, and convergent claims are kept separate throughout.

Keywords. Inverse Gamma function; Barnes G-function; Lambert W; transseries; series reversion; Hahn series; Stirling series; Lagrange–Bürmann inversion; Newton–Hensel lifting; exponentially improved asymptotics.

Contents

1 Introduction and principal formulas

The inversion of a rapidly growing special function has two logically distinct layers. The first layer inverts the dominant growth law. The second layer propagates every lower-order asymptotic term through the inverse map. For polynomial growth the first layer is algebraic; for an exponential it is a logarithm; for a phase of the form $x^p \log x$, the exact core inverse is a Lambert W-expression. Gamma and Barnes G are the first two canonical examples of the latter class.

The familiar leading estimate for inverse Gamma follows by solving Stirling's main term. What is less visible in that one-line calculation is the hierarchy of scales. Expanding Lambert W creates an outer series in powers of $1/\log\log y$, with polynomial dependence on $\log\log\log y$. A fixed shift, such as the $1/2$ in the optimal Gamma centering, is smaller than every term of that outer scale. The Bernoulli corrections are smaller again: they form a grid in powers of the inverse Lambert core, with coefficient functions that themselves have inverse-logarithmic expansions. Treating all these objects as if they were terms of one ordinary power series obscures both the algebra and the attainable error estimates.

The Barnes problem has one further feature. Its forward phase contains $-\frac{1}{12}\log z$. After normalization by the dominant $\frac{1}{2}z^2\log z$ phase, this term is small by z^{-2}, but its factor $\log z$ is of the same order as the core slope parameter. It therefore survives in the coefficientwise large-slope limit and creates an algebraic subseries inside the inverse transseries. This is the simplest example of the *resonant logarithmic correction* developed in Section 5.

1.1 Branch conventions

On the positive real axis, Γ has a unique minimum: indeed, $\psi'(x) = \sum_{n\geq 0}(n+x)^{-2} > 0$, while $\psi(x) \to -\infty$ as $x \to 0^+$ and $\psi(x) \to +\infty$ as $x \to +\infty$. Thus ψ has one zero and $\Gamma' = \Gamma\psi$ changes sign exactly once. Consequently, a large positive y has two positive preimages. We write $\Gamma_+^{-1}(y)$ for the branch tending to $+\infty$ and $\Gamma_0^{-1}(y)$ for the branch tending to 0^+. The Lambert-core transseries concerns Γ_+^{-1}; the second branch has an ordinary series in $1/y$, given in Section 9.

For Barnes G, we use $\mathcal{B}(y)$ for the eventual positive real inverse of $z \mapsto G(z+1)$:

$$G(\mathcal{B}(y)+1) = y, \qquad \mathcal{B}(y) \longrightarrow +\infty.$$

Thus the corresponding branch of the inverse of G itself is

$$G_+^{-1}(y) = 1 + \mathcal{B}(y).$$

This convention removes a persistent off-by-one shift from the asymptotic formulas.

1.2 The answer at a glance

Main result

Let $B_n(x)$ denote a Bernoulli polynomial and $B_n = B_n(0)$.
Upper inverse Gamma. Define

$$R_\Gamma = \log y - \frac{1}{2}\log(2\pi), \quad w = \mathrm{W}_0(R_\Gamma/\mathrm{e}), \quad T = \frac{R_\Gamma}{w}, \quad \Lambda = \log T = w + 1. \tag{1}$$

Then, for each fixed $N \geq 0$,

$$\Gamma_+^{-1}(y) = \frac{1}{2} + T + \sum_{n=1}^{N} \frac{d_{2n-1}(\Lambda)}{T^{2n-1}} + O\!\left(\frac{1}{T^{2N+1}\Lambda}\right), \qquad y \to +\infty, \tag{2}$$

where the d_{2n-1} are generated by Equation (62). In particular,

$$d_1(\Lambda) = \frac{1}{24\Lambda}, \tag{3}$$

$$d_3(\Lambda) = -\frac{14\Lambda^2 + 10\Lambda + 5}{5760\Lambda^3}, \tag{4}$$

$$d_5(\Lambda) = \frac{2232\Lambda^4 + 1176\Lambda^3 + 714\Lambda^2 + 350\Lambda + 105}{2903040\Lambda^5}. \tag{5}$$

Upper inverse Barnes G. Put

$$\alpha = \frac{1}{2}\log(2\pi), \quad C_G = \zeta'(-1), \quad R_G = \log y - C_G, \tag{6}$$

$$v = \mathrm{W}_0(4R_G/\mathrm{e}^3), \qquad U = 2\sqrt{R_G/v}, \qquad \lambda = \frac{v+1}{2} = \log U - 1. \tag{7}$$

Then, for each fixed $N \geq 0$,

$$\mathcal{B}(y) = U + \sum_{n=0}^{N} \frac{b_n(\lambda)}{U^n} + O(U^{-N-1}), \qquad y \to +\infty, \tag{8}$$

with coefficients generated by Equation (101). The first three are

$$b_0(\lambda) = -\frac{\alpha}{\lambda}, \tag{9}$$

$$b_1(\lambda) = \frac{6\alpha^2\lambda - 6\alpha^2 + \lambda^3 + \lambda^2}{12\lambda^3}, \tag{10}$$

$$b_2(\lambda) = \frac{\alpha(8\alpha^2\lambda - 6\alpha^2 + \lambda^2)}{12\lambda^5}. \tag{11}$$

Finally, $G_+^{-1}(y) = 1 + \mathcal{B}(y)$.

1.3 Relation to earlier work

The forward asymptotic expansions used below are classical. We take the Gamma form from the NIST Digital Library of Mathematical Functions [4] and the Barnes form from DLMF and Nemes [5, 10]. Corless, Gonnet, Hare, Jeffrey, and Knuth provide the standard analytic and asymptotic framework for Lambert W [3]. Jeffrey and Watt discuss branches of the complex inverse Gamma function and record the leading Lambert approximation [8]. Gerhold, Hubalek, and Tomovski prove the inverse-Gamma expansion through the T^{-3} block in notation equivalent to ours [7]. The present derivation recovers those terms, continues them by a closed triangular recurrence, treats Barnes G in the same framework, and isolates the general support and resonance mechanisms.

No assertion of convergence is hidden in the symbol $\sim$. Stirling's Bernoulli series is generally divergent, so the inverse block expansions are Poincaré asymptotic transseries. A finite truncation is rigorous once the corresponding forward remainder is controlled; Section 6 explains that transfer. Exponentially improved forward expansions lead to deeper inverse sectors, but they do not make the untruncated Bernoulli series convergent.

Scale warning

There are three different operations that are easily conflated:

1. retaining Lambert W exactly and expanding only the lower-order phase corrections;

2. expanding Lambert W itself in iterated logarithms;
3. optimally truncating the divergent Bernoulli sector and then adding exponentially small terms.

They correspond to three nested asymptotic levels. An equality or error term valid at one level need not survive a reordering at another.

2 Exact inversion of the Lambert core

2.1 The model phase

Let

$$\Phi_0(x) = ax^p(\log x - \beta), \qquad a > 0, \quad p > 0, \tag{12}$$

and suppose first that all quantities are positive real and that the branch at large x is intended. The following elementary lemma is the organizing identity for the whole article.

Lemma 2.1 (Exact Lambert-core inverse). *Let $R > 0$, and define*

$$Z = \frac{pR}{a}\mathrm{e}^{-p\beta}, \qquad v = \mathrm{W}_0(Z), \qquad X = \mathrm{e}^{\beta}\mathrm{e}^{v/p} = \left(\frac{pR}{av}\right)^{1/p}. \tag{13}$$

Then $\Phi_0(X) = R$. Moreover,

$$\log X = \beta + \frac{v}{p}, \qquad \Phi_0'(X) = aX^{p-1}(v+1), \tag{14}$$

and

$$\frac{\mathrm{d}X}{\mathrm{d}R} = \frac{1}{aX^{p-1}(v+1)}. \tag{15}$$

Proof. Put $u = p(\log X - \beta)$. Equation $\Phi_0(X) = R$ is equivalent to

$$\frac{a}{p}\mathrm{e}^{p\beta}u\mathrm{e}^{u} = R,$$

or $u\mathrm{e}^u = Z$. Thus $u = v = \mathrm{W}_0(Z)$, which gives Equation (13). Differentiating $ax^p(\log x - \beta)$ and substituting $p(\log X - \beta) = v$ yields Equation (14); the inverse-function formula then gives Equation (15). ☐

The derivative in Equation (14) identifies the universal divisor $v+1$ that appears in every correction coefficient. It also explains why a small phase residual should be divided by $aX^{p-1}(v+1)$, not merely by the leading power X^{p-1}.

2.2 The outer iterated-logarithmic layer

For $|Z| \to \infty$ in a fixed sector avoiding the standard cuts, write

$$L_1 = \log Z, \qquad L_2 = \log L_1.$$

The classical expansion

$$\mathrm{W}(Z) \sim L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2(-2+L_2)}{2L_1^2} + \frac{L_2(6 - 9L_2 + 2L_2^2)}{6L_1^3} + \cdots \tag{16}$$

turns $X = (pR/(av))^{1/p}$ into a polynomial-logarithmic transseries in $1/L_1$. We deliberately keep $v = \mathrm{W}(Z)$ exact for most of the article. Doing so packages infinitely many outer logarithmic terms into one analytic quantity and exposes the much smaller correction grid generated by the remainder of the phase.

Define

$$q = X^{-1} = \mathrm{e}^{-\beta - v/p}. \tag{17}$$

As $v \to +\infty$,

$$q \prec v^{-N} \qquad \text{for every fixed } N > 0. \tag{18}$$

Thus powers of q are beyond all orders of the coefficient scale v^{-1}. The natural inverse object is consequently a two-level transseries: a Hahn series in q, whose coefficients are rational or formal Laurent series in v^{-1}.

3 Normalized displacement and the triangular inverse theorem

3.1 The exact normalized equation

Consider a perturbed phase

$$\Phi(x) = ax^p(\log x - \beta) + C + \Psi(x) \tag{19}$$

and the equation $\Phi(x) = Y$. Set $R = Y - C$, construct v, X, q by Equations (13) and (17), and write

$$x = X(1 + \epsilon). \tag{20}$$

After division by aX^p, the core displacement is

$$\mathcal{K}_{p,v}(\epsilon) = (1+\epsilon)^p \left(\frac{v}{p} + \log(1+\epsilon)\right) - \frac{v}{p}. \tag{21}$$

The normalized perturbation is

$$\mathcal{R}(q, v, \epsilon) = \frac{\Psi(X(1+\epsilon))}{aX^p}. \tag{22}$$

Hence the inverse problem is exactly

$$\mathcal{K}_{p,v}(\epsilon) + \mathcal{R}(q, v, \epsilon) = 0. \tag{23}$$

The Taylor coefficients of the universal kernel have a compact closed form:

$$\mathcal{K}_{p,v}(\epsilon) = \sum_{n \geq 1} \kappa_n(p, v)\epsilon^n, \qquad \kappa_n(p, v) = \frac{v}{p}\binom{p}{n} + \frac{\partial}{\partial p}\binom{p}{n}. \tag{24}$$

Indeed, $(1+\epsilon)^p \log(1+\epsilon) = \partial_p (1+\epsilon)^p$. In particular,

$$\kappa_1(p, v) = v + 1, \tag{25}$$

which is the normalized core slope.

3.2 Hahn-series setting

Let $S \subset \mathbb{R}_{\geq 0}$ be a well-ordered additive submonoid with $0 \in S$, and let $\mathscr{C}$ be a characteristic-zero coefficient field containing p, v and $(v+1)^{-1}$. Typical choices are $\mathbb{C}(v)$, $\mathbb{C}((v^{-1}))$, or a differential transseries field extending them. Write

$$\mathscr{C}[[q^S]] = \left\{ \sum_{\rho \in S} c_\rho q^\rho : \operatorname{supp}(c) \text{ is well ordered} \right\}$$

for the corresponding Hahn ring and

$$\mathfrak{m} = \{F \in \mathscr{C}[[q^S]] : \operatorname{ord}_q F > 0\}$$

for its positive-valuation ideal.

The condition that S be well ordered is not decorative. It ensures that at any prescribed exponent ρ, only finitely many decompositions of ρ can contribute to the coefficient of a product or composition under the finite-type hypotheses used below. For the two principal examples, $S = 2\mathbb{N}$ for shifted Gamma and $S = \mathbb{N}$ for Barnes G, so no transfinite bookkeeping is needed; the Hahn formulation states the closure principle that survives fractional powers and mixed grids.

Theorem 3.1 (Triangular Lambert-core inversion)**.** *Assume that*

$$\mathcal{R}(q, v, E) = \sum_{\substack{\rho \in S,\ \rho > 0 \\ m \geq 0}} r_{\rho,m}(v) q^\rho E^m \tag{26}$$

is locally finite at every q-exponent and belongs to $\mathfrak{m}\mathscr{C}[[E]]$. If $v+1$ is invertible in $\mathscr{C}$, then Equation (23) has a unique solution

$$\epsilon(q, v) = \sum_{\substack{\gamma \in S \\ \gamma > 0}} e_\gamma(v) q^\gamma \in \mathfrak{m}. \tag{27}$$

For every $\gamma > 0$, let $\epsilon_{<\gamma} = \sum_{0<\delta<\gamma} e_\delta q^\delta$. Then

$$e_\gamma(v) = -\frac{1}{v+1} [q^\gamma] \left(\sum_{n \geq 2} \kappa_n(p, v) \epsilon_{<\gamma}^n + \mathcal{R}(q, v, \epsilon_{<\gamma}) \right). \tag{28}$$

Proof. Because $\epsilon \in \mathfrak{m}$, every nonlinear power ϵ^n, $n \geq 2$, has strictly larger valuation than ϵ. Likewise, every monomial $q^\rho \epsilon^m$ in $\mathcal{R}$ has $\rho > 0$. Therefore the coefficient of q^γ in all terms of Equation (23) except $(v+1)\epsilon$ depends only on coefficients e_δ with $\delta < \gamma$. Local finiteness makes that coefficient a finite sum. Solving the coefficient equation gives Equation (28); transfinite induction over the well-ordered support constructs a solution.

For uniqueness, suppose two solutions first differ at exponent γ. All nonlinear and perturbed contributions at that exponent depend only on lower coefficients and therefore agree. The difference of the two equations is $(v+1)(e_\gamma - \widetilde{e}_\gamma) = 0$, so the coefficients agree, a contradiction. □

Corollary 3.2 (Support semigroup)**.** *If the q-support of $\mathcal{R}(q, v, 0)$ is contained in a set $A \subset \mathbb{R}_{>0}$, and all E-dependent coefficients of $\mathcal{R}$ use the same base exponents, then the support of ϵ is contained in the additive semigroup generated by A.*

Proof. The recurrence combines previously occurring exponents only by addition and adds base exponents from A. Induction gives the claim. □

3.3 A denominator bound on an integral grid

The ubiquitous powers of the slope divisor have a simple combinatorial origin.

Proposition 3.3 (Universal slope-denominator bound)**.** *Suppose $S = \mathbb{N}$, the coefficients of $\mathcal{R}$ lie in a polynomial ring $A[v]$, and no coefficient contains $(v+1)^{-1}$. If $\epsilon = \sum_{n \geq 1} e_n q^n$, then*

$$(v+1)^{2n-1} e_n \in A[v] \qquad (n \geq 1). \tag{29}$$

The exponent $2n-1$ is an upper bound; cancellations may lower it.

Proof. The assertion is immediate for $n = 1$. Assume it for every lower index. A coefficient of q^n in a nonlinear product $e_{j_1} \cdots e_{j_m}$, where $m \geq 2$ and $j_1 + \cdots + j_m = n$, has denominator exponent at most

$$\sum_{r=1}^{m}(2j_r - 1) = 2n - m \leq 2n - 2.$$

After the final division by $v + 1$ in Equation (28), the exponent is at most $2n - 1$. A monomial from $\mathcal{R}$ carries an additional positive base exponent q^ρ, so its indices sum to $n - \rho < n$ and gives a strictly smaller bound. This closes the induction. □

3.4 Formal versus analytic meaning

Theorem 3.1 is a formal implicit-function theorem. To obtain an asymptotic theorem for an analytic function, one also needs:

1. a forward Poincaré expansion for Ψ, with a remainder uniform in the sector or interval under consideration;
2. eventual nonvanishing of $\Phi'(x)$;
3. a residual-to-root estimate, supplied in Section 6.

The formal recurrence then computes a candidate truncation, while the analytic hypotheses prove that the exact inverse differs from it by the stated next scale. This separation prevents a common error: formal existence in a Hahn ring does not imply convergence of the resulting infinite series.

4 Equivalent reversion mechanisms

The normalized displacement recurrence is the most efficient route to the explicit coefficients in this article. Two complementary formulations are valuable both conceptually and computationally: a reduction to near-identity reversion in the phase coordinate, and a perturbation expansion in a formal coupling parameter.

4.1 Phase-coordinate reduction and Lagrange–Bürmann inversion

Let $h = \Phi_0^{-1}$ denote the chosen Lambert-core inverse. Introduce the phase coordinate $u = \Phi_0(x)$. Then

$$\Phi(h(u)) = u + C + A(u), \qquad A(u) = \Psi(h(u)). \tag{30}$$

After replacing Y by $R = Y - C$, inversion of Φ factors as

$$\Phi^{-1}(Y) = h\big((I + A)^{-1}(R)\big), \qquad (I + A)(u) = u + A(u). \tag{31}$$

Thus every Lambert-core problem can be reduced to the near-identity reversion calculus already natural for polynomial-logarithmic transseries.

Theorem 4.1 (Lagrange–Bürmann phase reversion)**.** *Let $\mathscr{D}$ be a differential transseries algebra in which the following family is locally finite in the chosen valuation. Then*

$$(I + A)^{-1}(R) = R + \sum_{n\geq 1} \frac{(-1)^n}{n!} \left(\frac{\mathrm{d}}{\mathrm{d}R}\right)^{n-1} A(R)^n. \tag{32}$$

Consequently,

$$\Phi^{-1}(Y) = h\left(R + \sum_{n\geq 1} \frac{(-1)^n}{n!} D_R^{n-1} A(R)^n\right). \tag{33}$$

Proof. The classical Lagrange–Bürmann identity for the inverse of $u \mapsto u + A(u)$ gives Equation (32). In a formal transseries algebra, the only additional issue is summability: at every prescribed monomial, only finitely many summands may contribute. This is the stated local-finiteness hypothesis. Composition with the already defined core inverse h gives Equation (33). □

Remark 4.2. Formula Equation (32) is universal but not always the best coefficient engine. Repeated differentiation expands expressions that the normalized recurrence keeps factored. It is most useful for structural proofs, for deriving the first few perturbative terms, and for importing a pre-existing near-identity transseries calculus.

4.2 Perturbation coefficients in closed differential form

Insert a formal parameter τ and solve

$$\Phi_0(x) + \tau\Psi(x) = R, \qquad x = X + \tau x_1 + \tau^2 x_2 + \tau^3 x_3 + \cdots . \tag{34}$$

Write, at $x = X$,

$$P = \Phi_0'(X), \quad Q = \Phi_0''(X), \quad S = \Phi_0'''(X), \qquad \Psi_j = \Psi^{(j)}(X).$$

Direct Taylor expansion gives

$$x_1 = -\frac{\Psi_0}{P}, \tag{35}$$

$$x_2 = \frac{\Psi_0\Psi_1}{P^2} - \frac{Q\Psi_0^2}{2P^3}, \tag{36}$$

$$x_3 = -\frac{\Psi_0\Psi_1^2}{P^3} - \frac{\Psi_0^2\Psi_2}{2P^3} + \frac{3Q\Psi_0^2\Psi_1}{2P^4} + \frac{S\Psi_0^3}{6P^4} - \frac{Q^2\Psi_0^3}{2P^5}. \tag{37}$$

These identities are useful when Ψ is treated as one composite object. The triangular q-recurrence is finer: it separates the different scales inside Ψ and computes only the coefficient requested at each step.

Derivation. Substitute Equation (34) into the left-hand side and equate powers of τ. At first order one obtains $Px_1 + \Psi_0 = 0$. At second order,

$$Px_2 + \frac{1}{2}Qx_1^2 + \Psi_1 x_1 = 0.$$

At third order,

$$Px_3 + Qx_1x_2 + \frac{1}{6}Sx_1^3 + \Psi_1 x_2 + \frac{1}{2}\Psi_2 x_1^2 = 0.$$

Solving successively and simplifying yields Equations (35) to (37). □

4.3 Newton–Hensel lifting

Define

$$F(\epsilon) = \mathcal{K}_{p,v}(\epsilon) + \mathcal{R}(q, v, \epsilon).$$

Formal Newton iteration is

$$\epsilon_{\text{new}} = \epsilon - \frac{F(\epsilon)}{\partial_\epsilon F(\epsilon)}. \tag{38}$$

Since $\partial_\epsilon F(0) = v + 1 + O(q)$ is a unit, Newton lifting doubles the known q-valuation under the usual complete-ring hypotheses. A triangular step is preferable when coefficients are wanted sequentially; Newton is preferable for high-order truncated-series arithmetic because one inversion of a unit replaces many scalar recurrences.

5 Resonant logarithmic corrections

The large parameter v appears in two places: in the core kernel and in the coefficients obtained by evaluating powers of $\log x$. Usually the latter are lower order after division by the core slope. A term containing exactly one factor of $\log x$, however, may survive coefficientwise as $v \to \infty$. Barnes G exhibits this phenomenon.

5.1 The coefficientwise limiting equation

Suppose the normalized perturbation has an expansion

$$\mathcal{R}(q, v, E) = vH(q, E) + R_0(q, v, E), \qquad v^{-1}R_0 \longrightarrow 0 \tag{39}$$

coefficientwise in the Hahn topology. Divide Equation (23) by v. Because

$$\frac{1}{v}\mathcal{K}_{p,v}(E) \longrightarrow \frac{(1+E)^p - 1}{p}, \tag{40}$$

the limiting correction $E_0(q)$, when it exists, satisfies

$$\frac{(1+E_0)^p - 1}{p} + H(q, E_0) = 0, \qquad E_0(0) = 0. \tag{41}$$

Theorem 5.1 (Resonant coefficient limit)**.** *Under the hypotheses of Theorem 3.1, assume additionally Equation (39) and suppose that the derivative of the left-hand side of Equation (41) at $E_0 = 0, q = 0$ is a unit. Then the coefficients of the unique solution $\epsilon(q, v)$ tend, as $v \to \infty$, to the coefficients of the unique small solution $E_0(q)$ of Equation (41).*

Proof. Apply the triangular recurrence coefficient by coefficient after division by v. At each exponent, the coefficient equation has a limiting nonzero linear divisor and depends only on already determined lower coefficients. Induction therefore commutes with the coefficientwise limit. □

For a perturbation monomial $cx^{p-\rho}\log x$, substitution $x = X(1+E)$ gives the leading normalized contribution

$$\frac{c}{ap}vq^{\rho}(1+E)^{p-\rho}, \tag{42}$$

so such monomials are precisely the elementary sources of resonance. Terms without $\log x$ disappear from the limiting equation, while higher powers of $\log x$ require a modified scaling and may change the dominant core rather than merely perturb it.

6 Residuals, rigorous errors, and exponential sectors

6.1 Residual-to-root transfer

A symbolic transseries truncation becomes an analytic approximation only after its residual is transferred through the inverse slope.

Theorem 6.1 (Residual certificate)**.** *Let $\Phi \in C^2(I)$ be real and strictly monotone on an interval I, let $x \in I$ solve $\Phi(x) = Y$, and let $\widehat{x} \in I$. If $|\Phi'(t)| \geq m > 0$ between x and $\widehat{x}$, then*

$$|x - \widehat{x}| \leq \frac{|\Phi(\widehat{x}) - Y|}{m}. \tag{43}$$

If the residual $\delta = \Phi(\widehat{x}) - Y$ tends to zero and the relevant derivatives have regular asymptotics, then

$$x - \widehat{x} = -\frac{\delta}{\Phi'(\widehat{x})} + O\left(\frac{\Phi''(\widehat{x})\delta^2}{\Phi'(\widehat{x})^3}\right). \tag{44}$$

Proof. The first assertion is the mean-value theorem. For the second, write $x = \widehat{x} + h$, Taylor-expand $0 = \delta + \Phi'(\widehat{x})h + \frac{1}{2}\Phi''(\xi)h^2$, and use the first bound to control h. □

For Equation (19), provided $\Psi'(X) = o(X^{p-1}v)$,

$$\Phi'(X) \sim aX^{p-1}(v+1). \tag{45}$$

Thus a forward phase residual δ produces an inverse error of leading size

$$-\frac{\delta}{aX^{p-1}(v+1)}. \tag{46}$$

This single division explains the $1/\Lambda$ factor in the Gamma remainder and the $1/(U\lambda)$ response of the Barnes inverse to a phase error.

6.2 Transport of exponentially small sectors

Suppose an exponentially improved forward phase contains a sector

$$E_\sigma(x) = S_\sigma(x)\exp(-\sigma x^\mu), \qquad \Re(\sigma x^\mu) > 0. \tag{47}$$

At first order, its inverse contribution is

$$\Delta x_\sigma \sim -\frac{S_\sigma(X)\exp(-\sigma X^\mu)}{aX^{p-1}(v+1)}. \tag{48}$$

Products and substitutions of these sectors generate the deeper transseries levels in the usual way. Stokes multipliers are transported by the same rule: evaluate the forward sector at the algebraic-logarithmic inverse and divide by the inverse slope, then iterate if higher accuracy is required.

For Gamma and Barnes G, explicit error bounds and exponentially improved forward expansions are available in the literature [9, 10]. The present article does not rederive their terminant-function structure. It identifies the exact algebraic operation needed to convert any chosen forward hyperasymptotic truncation into an inverse one.

Scale warning

An optimally truncated Bernoulli series may have an exponentially small remainder, but the formal infinite Bernoulli series remains divergent. The exponential sector belongs after optimal truncation; it must not be appended to an unqualified equality with the divergent series.

7 Complex asymptotic branches

The positive real formulas use $\log y$ and W_0. Complex inversion requires two branch choices. Let

$$Y_m = \operatorname{Log} y + 2\pi i m - C, \qquad m \in \mathbb{Z}, \tag{49}$$

where Log is a fixed principal determination at the target. For every Lambert branch index k, define

$$v_{m,k} = \mathrm{W}_k\left(\frac{pY_m}{a}\mathrm{e}^{-p\beta}\right), \qquad X_{m,k} = \exp\left(\beta + \frac{v_{m,k}}{p}\right). \tag{50}$$

The exponential form is preferable to an ambiguous p-th root: it keeps the logarithm and Lambert branches compatible. Whenever the forward phase has a uniform asymptotic expansion in a sector containing $X_{m,k}$, the same normalized recurrence gives a formal inverse branch based at that core. A Rouché argument or a Newton contraction then identifies an actual local inverse sheet for sufficiently large $|Y_m|$.

For Gamma, the half-shifted variable and $p = 1$ leave the pair (m, k) as the asymptotic sheet labels. For Barnes G, $p = 2$ rotates the core by approximately half the change in the Lambert phase. Global branch topology is more intricate because of poles or zeros of the original special function; the construction here is an asymptotic local classification, not a claim that the labels are globally one-to-one.

8 The upper inverse Gamma transseries

8.1 Why the half-shift is the correct normal form

The standard Stirling expansion may be written, uniformly in sectors bounded away from the negative real axis, as

$$\log\Gamma(z+h) \sim \left(z+h-\frac{1}{2}\right)\log z - z + \frac{1}{2}\log(2\pi) + \sum_{k=2}^{\infty}\frac{(-1)^k B_k(h)}{k(k-1)z^{k-1}}. \tag{51}$$

Set $h = 1/2$ and write

$$t = x - \frac{1}{2}. \tag{52}$$

Because $B_{2k+1}(1/2) = 0$, every even inverse power disappears and

$$\log\Gamma\left(t+\frac{1}{2}\right) \sim t(\log t - 1) + \frac{1}{2}\log(2\pi) + \sum_{k\geq 1}\frac{a_k}{t^{2k-1}}, \tag{53}$$

where

$$a_k = \frac{B_{2k}(1/2)}{2k(2k-1)} = \frac{(2^{1-2k}-1)B_{2k}}{2k(2k-1)}. \tag{54}$$

The first values are

$$a_1 = -\frac{1}{24}, \quad a_2 = \frac{7}{2880}, \quad a_3 = -\frac{31}{40320}, \quad a_4 = \frac{127}{215040}, \quad a_5 = -\frac{511}{608256}. \tag{55}$$

The half-shift is more than a cosmetic acceleration. It changes the support of the normalized perturbation from a full integer grid to the even grid $2\mathbb{N}$. Consequently, the displacement $t - T$ contains only odd inverse powers of the Lambert core. This parity is exact to all algebraic orders.

8.2 Core variables and normalized equation

Let $y \to +\infty$ and define

$$R = \log y - \frac{1}{2}\log(2\pi), \qquad w = \mathrm{W}_0(R/\mathrm{e}), \qquad T = \frac{R}{w} = \mathrm{e}^{w+1}, \qquad \Lambda = \log T = w + 1. \tag{56}$$

Then

$$T(\log T - 1) = R, \qquad \frac{\mathrm{d}}{\mathrm{d}T}\big(T(\log T - 1)\big) = \Lambda. \tag{57}$$

Put $q = T^{-1}$ and

$$t = T(1+\epsilon), \qquad \epsilon = \sum_{n\geq 1} e_n(\Lambda) q^{2n}. \tag{58}$$

Subtract the core equation from Equation (53) and divide by T. The exact formal equation is

$$\Lambda\epsilon + (1+\epsilon)\log(1+\epsilon) - \epsilon + \sum_{k\geq 1} a_k q^{2k}(1+\epsilon)^{-(2k-1)} = 0. \tag{59}$$

Since $t - T = T\epsilon$, define

$$d_{2n-1}(\Lambda) = e_n(\Lambda). \tag{60}$$

Theorem 8.1 (All-orders upper inverse Gamma expansion). *Let Γ_+^{-1} be the positive real inverse branch tending to infinity. For every fixed $N \geq 0$,*

$$\Gamma_+^{-1}(y) = \frac{1}{2} + T + \sum_{n=1}^{N} \frac{d_{2n-1}(\Lambda)}{T^{2n-1}} + O\left(\frac{1}{T^{2N+1}\Lambda}\right), \qquad y \to +\infty, \tag{61}$$

where T, Λ are given by Equation (56). The coefficients are uniquely determined by $d_{2n-1} = e_n$ and

$$e_n(\Lambda) = -\frac{1}{\Lambda}\left[q^{2n}\right]\left[(1+E_{n-1})\log(1+E_{n-1}) - E_{n-1} + \sum_{k=1}^{n} a_k q^{2k}(1+E_{n-1})^{-(2k-1)}\right], \tag{62}$$

with

$$E_{n-1} = \sum_{j=1}^{n-1} e_j(\Lambda) q^{2j}. \tag{63}$$

Moreover,

$$d_{2n-1}(\Lambda) \in \Lambda^{-1}\mathbb{Q}[\Lambda^{-1}], \qquad \deg_{\Lambda^{-1}} d_{2n-1} \leq 2n-1, \tag{64}$$

and

$$d_{2n-1}(\Lambda) = -\frac{a_n}{\Lambda} + O(\Lambda^{-2}). \tag{65}$$

Proof. Equation (59) is the specialization of Equation (23) with $p = a = \beta = 1$ and support $2\mathbb{N}$. The coefficient of ϵ is Λ, so Theorem 3.1 gives existence, uniqueness, and Equation (62).

We prove Equation (64) by induction. The nonlinear kernel $(1+E)\log(1+E) - E$ starts with $E^2/2$ and has rational coefficients. The binomial expansions in the Bernoulli sector also have rational coefficients. If the lower e_j are polynomials in Λ^{-1} with zero constant term and degree at most $2j-1$, then every coefficient inside the brackets in Equation (62) is a polynomial in Λ^{-1} of degree at most $2n-2$. Division by Λ gives the stated degree bound and zero constant term. At exponent q^{2n}, the only contribution independent of lower e_j is $a_n q^{2n}$, which proves Equation (65).

For the analytic remainder, use the duplication identity

$$\log\Gamma\left(t + \frac{1}{2}\right) = \log\Gamma(2t) - \log\Gamma(t) + (1-2t)\log 2 + \frac{1}{2}\log\pi.$$

Apply the standard remainder bound separately to the two ordinary positive-real Stirling expansions. For every fixed M, this shows that Equation (53) truncated after $a_M t^{-(2M-1)}$ has remainder $O(t^{-(2M+1)})$. Choose $M = N+1$. Substitution of the formal inverse through $e_N q^{2N}$ then leaves a forward phase residual $O(T^{-(2N+1)})$. The inverse phase slope is $\Lambda + O(T^{-2})$. Applying Theorem 6.1 divides the phase residual by Λ, yielding the error in Equation (61). □

8.3 Explicit correction blocks

The first five nonzero blocks are

$$d_1 = \frac{1}{24\Lambda}, \tag{66}$$

$$d_3 = -\frac{14\Lambda^2 + 10\Lambda + 5}{5760\Lambda^3}, \tag{67}$$

$$d_5 = \frac{2232\Lambda^4 + 1176\Lambda^3 + 714\Lambda^2 + 350\Lambda + 105}{2903040\Lambda^5}, \tag{68}$$

$$\begin{aligned} d_7 = -\frac{1}{1393459200\Lambda^7}\big(&822960\Lambda^6 + 292536\Lambda^5 + 136956\Lambda^4 \\ &+ 68040\Lambda^3 + 32970\Lambda^2 + 12250\Lambda + 2625\big), \end{aligned} \tag{69}$$

$$\begin{aligned} d_9 = \frac{1}{367873228800\Lambda^9}\big(&309052800\Lambda^8 + 77920128\Lambda^7 + 26024592\Lambda^6 \\ &+ 10019592\Lambda^5 + 4516028\Lambda^4 + 2023560\Lambda^3 + 812350\Lambda^2 \\ &+ 242550\Lambda + 40425\big). \end{aligned} \tag{70}$$

Thus, through the first three correction blocks,

$$\begin{aligned} \Gamma_+^{-1}(y) = \frac{1}{2} + T + \frac{1}{24T\Lambda} - \frac{14\Lambda^2 + 10\Lambda + 5}{5760T^3\Lambda^3} \\ + \frac{2232\Lambda^4 + 1176\Lambda^3 + 714\Lambda^2 + 350\Lambda + 105}{2903040T^5\Lambda^5} + O\left(\frac{1}{T^7\Lambda}\right). \end{aligned} \tag{71}$$

For comparison with expansions written as separate powers of $\log T$, the first blocks are

$$d_3 = -\frac{7}{2880\Lambda} - \frac{1}{576\Lambda^2} - \frac{1}{1152\Lambda^3}, \tag{72}$$

$$d_5 = \frac{31}{40320\Lambda} + \frac{7}{17280\Lambda^2} + \frac{17}{69120\Lambda^3} + \frac{5}{41472\Lambda^4} + \frac{1}{27648\Lambda^5}. \tag{73}$$

The expansion of Gerhold, Hubalek, and Tomovski is precisely Equation (71) through d_3/T^3, expressed in the separated form Equation (72) [7].

8.4 A hand derivation of the first two blocks

The recurrence is transparent at low order. Since

$$(1+E)\log(1+E) - E = \frac{1}{2}E^2 - \frac{1}{6}E^3 + \cdots,$$

the coefficient of q^2 in Equation (59) is $\Lambda e_1 + a_1$. Therefore

$$e_1 = -\frac{a_1}{\Lambda} = \frac{1}{24\Lambda}.$$

At order q^4, the contributions are

$$\Lambda e_2 + \frac{1}{2}e_1^2 - a_1 e_1 + a_2 = 0.$$

Substituting $a_1 = -1/24$, $a_2 = 7/2880$, and $e_1 = 1/(24\Lambda)$ yields Equation (67). Higher orders are no more conceptually difficult, but the number of partitions of the target exponent grows; truncated-series arithmetic is preferable to hand expansion.

8.5 Outer expansion in iterated logarithms

The Lambert core itself can be expanded without first writing the generic W-series. Let

$$L = \log R, \qquad \ell = \log L. \tag{74}$$

Writing $T = (R/L)S$ in $T(\log T - 1) = R$ gives the implicit equation

$$S\left(1 - \frac{\ell+1}{L} + \frac{\log S}{L}\right) = 1. \tag{75}$$

Ordinary formal reversion in L^{-1} yields

$$\begin{aligned} T = \frac{R}{L}\Bigg[1 + \frac{\ell+1}{L} + \frac{\ell(\ell+1)}{L^2} + \frac{(\ell-1)(\ell+1)(2\ell+1)}{2L^3} \\ + \frac{(\ell+1)(6\ell^3 - 8\ell^2 - 7\ell + 1)}{6L^4} \\ + \frac{(\ell+1)(12\ell^4 - 29\ell^3 - 17\ell^2 + 17\ell + 5)}{12L^5} + O\left(\frac{\ell^6}{L^6}\right)\Bigg]. \end{aligned} \tag{76}$$

Combining Equation (76) with Equation (61) gives a pure iterated-logarithmic expansion if desired. The ordering must be read lexicographically:

$$\frac{T}{L^N} \longrightarrow \infty \quad \text{for every fixed } N, \qquad 1 \gg \frac{1}{T\Lambda}. \tag{77}$$

Hence the shift $1/2$ is beyond all orders of the outer L^{-1} scale, and the Bernoulli inverse blocks are beyond that shift.

8.6 Numerical verification

To avoid numerical inversion error, the following test starts from an exact chosen abscissa x, forms $y = \Gamma(x)$, reconstructs T, Λ from y, and compares successive truncations with the original x. Let E_0 denote approximation minus exact value for $1/2 + T$, and let E_j use the same sign convention after including the first j correction blocks $d_1/T, d_3/T^3, \ldots$.

Table 1: Signed errors (approximation minus exact value) for the upper inverse Gamma expansion.

x	E_0	E_1	E_2	E_3	E_4
5	-6.1414×10^{-3}	2.8694×10^{-5}	-4.1862×10^{-7}	1.3452×10^{-8}	-8.2694×10^{-10}
10	-1.9470×10^{-3}	1.7430×10^{-6}	-5.7560×10^{-9}	4.4485×10^{-11}	-6.5578×10^{-13}
20	-7.1924×10^{-4}	1.4126×10^{-7}	-1.1183×10^{-10}	2.1113×10^{-13}	-7.5612×10^{-16}
100	-9.1031×10^{-5}	6.2869×10^{-10}	-1.9388×10^{-14}	1.4440×10^{-18}	-2.0231×10^{-22}

The clean reduction by several orders per block is typical before optimal truncation. At a fixed moderate x, sufficiently remote Bernoulli blocks will eventually grow because the underlying Stirling series is divergent.

9 The other positive inverse Gamma branch

For completeness, a large positive y also has a positive preimage tending to zero. This branch is governed by a local pole, not by the Lambert core. Put $s = 1/y$. Since

$$s = \frac{x}{\Gamma(1+x)}, \tag{78}$$

Lagrange inversion gives the convergent local series

$$\Gamma_0^{-1}(y) = \sum_{n\ge 1} \frac{s^n}{n!} \frac{\mathrm{d}^{n-1}}{\mathrm{d}x^{n-1}} \Gamma(1+x)^n \bigg|_{x=0}, \qquad s = \frac{1}{y}. \tag{79}$$

Using

$$\log\Gamma(1+x) = -\gamma x + \sum_{m\ge 2} \frac{(-1)^m \zeta(m)}{m} x^m, \qquad |x| < 1, \tag{80}$$

we find

$$\begin{aligned}\Gamma_0^{-1}(y) = {}& \frac{1}{y} - \frac{\gamma}{y^2} + \left(\frac{3\gamma^2}{2} + \frac{\pi^2}{12}\right)\frac{1}{y^3} \\ & - \left(\frac{8\gamma^3}{3} + \frac{\gamma\pi^2}{3} + \frac{\zeta(3)}{3}\right)\frac{1}{y^4} \\ & + \left(\frac{125\gamma^4}{24} + \frac{25\gamma^2\pi^2}{24} + \frac{5\gamma\zeta(3)}{3} + \frac{29\pi^4}{1440}\right)\frac{1}{y^5} + O(y^{-6}).\end{aligned} \tag{81}$$

More generally, every pole at $-n$ gives a local inverse branch with leading term

$$x = -n + \frac{(-1)^n}{n!\,y} + O(y^{-2}). \tag{82}$$

These pole-centered branches belong to ordinary local series reversion and are algebraically separate from the large branch analyzed in Section 8.

10 The inverse Barnes G-transseries

10.1 Forward phase in a centered variable

Barnes' G-function is characterized by

$$G(z+1) = \Gamma(z)G(z), \qquad G(1) = 1. \tag{83}$$

For inversion on the large positive real axis, write the argument as $z+1$ from the outset. The classical forward expansion is

$$\begin{aligned} \log G(z+1) \sim{}& \frac{1}{2}z^2\log z - \frac{3}{4}z^2 + \frac{1}{2}\log(2\pi)\,z - \frac{1}{12}\log z + \zeta'(-1) \\ &+ \sum_{k\ge1} \frac{B_{2k+2}}{4k(k+1)z^{2k}}, \qquad z\to\infty, \end{aligned} \tag{84}$$

valid in every fixed sector avoiding the negative real axis [5, 10]. Define

$$\alpha = \frac{1}{2}\log(2\pi), \qquad b = -\frac{1}{12}, \qquad C_G = \zeta'(-1) = \frac{1}{12} - \log A, \tag{85}$$

where A is the Glaisher–Kinkelin constant, and

$$c_k = \frac{B_{2k+2}}{4k(k+1)}. \tag{86}$$

Then

$$c_1 = -\frac{1}{240}, \qquad c_2 = \frac{1}{1008}, \qquad c_3 = -\frac{1}{1440}, \qquad c_4 = \frac{1}{1056}. \tag{87}$$

In the notation of the general model,

$$a = \frac{1}{2}, \qquad p = 2, \qquad \beta = \frac{3}{2}, \tag{88}$$

and the perturbation is

$$\Psi(z) = \alpha z + b\log z + \sum_{k\ge1} c_k z^{-2k}. \tag{89}$$

10.2 Exact core variables

Let $G(z+1) = y$ and set

$$R = \log y - C_G. \tag{90}$$

The core equation

$$\frac{1}{2}U^2\left(\log U - \frac{3}{2}\right) = R \tag{91}$$

is solved exactly by

$$v = \mathrm{W}_0(4R/\mathrm{e}^3), \qquad U = 2\sqrt{\frac{R}{v}} = \mathrm{e}^{(v+3)/2}, \qquad \lambda = \log U - 1 = \frac{v+1}{2}. \tag{92}$$

The core slope is

$$\frac{\mathrm{d}}{\mathrm{d}U}\left[\frac{1}{2}U^2\left(\log U-\frac{3}{2}\right)\right]=U(\log U-1)=U\lambda. \tag{93}$$

Put $q=U^{-1}$ and

$$z=U(1+\epsilon),\qquad \epsilon=\sum_{n\geq 1}e_n(\lambda)q^n. \tag{94}$$

After subtraction of Equation (91) and division by $U^2/2$, the exact normalized equation is

$$\begin{aligned}0&=\mathcal{K}_\lambda(\epsilon)+2\alpha q(1+\epsilon)+2bq^2\big(\lambda+1+\log(1+\epsilon)\big)\\&\quad+2\sum_{k\geq 1}c_k q^{2k+2}(1+\epsilon)^{-2k},\end{aligned} \tag{95}$$

where

$$\mathcal{K}_\lambda(E)=(1+E)^2\left(\lambda-\frac{1}{2}+\log(1+E)\right)-\left(\lambda-\frac{1}{2}\right). \tag{96}$$

Its linear part is

$$\mathcal{K}_\lambda(E)=2\lambda E+O(E^2). \tag{97}$$

Since

$$z=U+e_1+e_2U^{-1}+e_3U^{-2}+\cdots,$$

we set

$$b_n(\lambda)=e_{n+1}(\lambda),\qquad n\geq 0. \tag{98}$$

Theorem 10.1 (All-orders inverse Barnes expansion). *Let $\mathcal{B}(y)$ be the eventual positive solution of $G(z+1)=y$. For every fixed $N\geq 0$,*

$$\mathcal{B}(y)=U+\sum_{n=0}^{N}\frac{b_n(\lambda)}{U^n}+O(U^{-N-1}),\qquad y\to+\infty, \tag{99}$$

where U,λ are given by Equation (92). Define

$$E_{n-1}=\sum_{j=1}^{n-1}e_j(\lambda)q^j. \tag{100}$$

Then e_n, and hence every b_n, is uniquely determined by

$$\begin{aligned}e_n(\lambda)=-\frac{1}{2\lambda}[q^n]\Big[&\mathcal{K}_\lambda(E_{n-1})-2\lambda E_{n-1}+2\alpha q(1+E_{n-1})\\&+2bq^2\big(\lambda+1+\log(1+E_{n-1})\big)\\&+2\sum_{1\leq k\leq (n-2)/2}c_kq^{2k+2}(1+E_{n-1})^{-2k}\Big].\end{aligned} \tag{101}$$

The last sum is empty when $n<4$. The coefficients satisfy

$$e_n\in\mathbb{Q}[\alpha,\lambda,\lambda^{-1}],\qquad e_n(-\alpha)=(-1)^n e_n(\alpha), \tag{102}$$

and are bounded coefficient functions as $\lambda\to+\infty$.

Proof. Equation (95) has support $\mathbb{N}$ and linear divisor 2λ. The general triangular theorem gives Equation (101). The coefficient ring follows by induction from the rational Taylor coefficients of the logarithm and binomial powers. The transformation

$$(q, \alpha, E) \longmapsto (-q, -\alpha, E) \tag{103}$$

leaves Equation (95) invariant. Uniqueness therefore implies $\epsilon(-q, -\alpha) = \epsilon(q, \alpha)$, which is equivalent to the parity assertion.

To prove boundedness, divide Equation (95) by λ. The only coefficient growing linearly with λ, besides the core kernel, is the constant part of the bq^2-term. The limiting equation has a small formal solution with bounded coefficients; Theorem 5.1 then gives coefficientwise convergence. The forward Barnes remainder can be truncated to any prescribed algebraic order. Its transfer through the slope $U\lambda(1 + o(1))$, together with boundedness of the next coefficient block, gives the error in Equation (99). □

10.3 Explicit Barnes coefficient blocks

The first coefficients are

$$b_0 = -\frac{\alpha}{\lambda}, \tag{104}$$

$$b_1 = \frac{6\alpha^2\lambda - 6\alpha^2 + \lambda^3 + \lambda^2}{12\lambda^3}, \tag{105}$$

$$b_2 = \frac{\alpha(8\alpha^2\lambda - 6\alpha^2 + \lambda^2)}{12\lambda^5}. \tag{106}$$

For the next two, define

$$\begin{aligned} P_3(\lambda, \alpha) = {} & 180\alpha^4\lambda^3 + 120\alpha^4\lambda^2 - 1500\alpha^4\lambda + 900\alpha^4 \\ & + 60\alpha^2\lambda^5 + 120\alpha^2\lambda^4 + 60\alpha^2\lambda^3 - 180\alpha^2\lambda^2 \\ & + 5\lambda^7 - \lambda^6 + 5\lambda^5 + 5\lambda^4, \end{aligned} \tag{107}$$

$$\begin{aligned} P_4(\lambda, \alpha) = {} & 576\alpha^4\lambda^3 + 480\alpha^4\lambda^2 - 2520\alpha^4\lambda + 1260\alpha^4 \\ & + 80\alpha^2\lambda^5 + 280\alpha^2\lambda^4 + 200\alpha^2\lambda^3 - 300\alpha^2\lambda^2 \\ & - 22\lambda^7 - 11\lambda^6 + 10\lambda^5 + 15\lambda^4. \end{aligned} \tag{108}$$

Then

$$b_3 = -\frac{P_3(\lambda, \alpha)}{1440\lambda^7}, \qquad b_4 = -\frac{\alpha P_4(\lambda, \alpha)}{1440\lambda^9}. \tag{109}$$

The longer coefficient b_5 is listed in Section B. The expansion through U^{-2} is therefore

$$\begin{aligned} \mathcal{B}(y) = {} & U - \frac{\alpha}{\lambda} + \frac{6\alpha^2\lambda - 6\alpha^2 + \lambda^3 + \lambda^2}{12\lambda^3 U} \\ & + \frac{\alpha(8\alpha^2\lambda - 6\alpha^2 + \lambda^2)}{12\lambda^5 U^2} + O(U^{-3}). \end{aligned} \tag{110}$$

10.4 Resonance and the limiting algebraic subseries

The resonance is completely explicit. Divide Equation (95) by 2λ and let $\lambda \to +\infty$ coefficientwise. The αq-term, all c_k-terms, and the nonconstant part of the logarithmic term vanish. Since

$$\frac{\mathcal{K}_\lambda(E)}{2\lambda} \longrightarrow \frac{(1+E)^2 - 1}{2},$$

the limiting series $E_0(q)$ satisfies

$$\frac{(1+E_0)^2-1}{2}+bq^2=0. \tag{111}$$

The branch with $E_0(0)=0$ is

$$E_0(q)=\sqrt{1-2bq^2}-1=\sqrt{1+\frac{q^2}{6}}-1. \tag{112}$$

Therefore

$$\lim_{\lambda\to\infty} e_{2m+1}(\lambda)=0, \qquad \lim_{\lambda\to\infty} e_{2m}(\lambda)=\binom{1/2}{m}6^{-m}. \tag{113}$$

Equivalently,

$$\lim_{\lambda\to\infty} b_{2m}(\lambda)=0, \qquad \lim_{\lambda\to\infty} b_{2m-1}(\lambda)=\binom{1/2}{m}6^{-m}. \tag{114}$$

The first limiting constants are

$$b_1\to\frac{1}{12}, \qquad b_3\to-\frac{1}{288}, \qquad b_5\to\frac{1}{3456}. \tag{115}$$

This coefficientwise algebraic limit is invisible if the forward $-\frac{1}{12}\log z$ term is treated as an undifferentiated small error. It is also a useful check on symbolic computations: every odd inverse-power block must approach the corresponding binomial coefficient, while every even block must vanish.

10.5 Outer iterated-logarithmic expansion

Let

$$Z=\frac{4R}{\mathrm{e}^3}, \qquad L_1=\log Z, \qquad L_2=\log L_1. \tag{116}$$

Expanding $v=\mathrm{W}_0(Z)$ and then $v^{-1/2}$ gives

$$\begin{aligned} U=2\sqrt{\frac{R}{L_1}}\Bigg[1&+\frac{L_2}{2L_1}+\frac{L_2(3L_2-4)}{8L_1^2}+\frac{L_2(5L_2^2-16L_2+8)}{16L_1^3}\\ &+\frac{L_2(105L_2^3-568L_2^2+720L_2-192)}{384L_1^4}+O\bigg(\frac{L_2^5}{L_1^5}\bigg)\Bigg]. \end{aligned} \tag{117}$$

For the inverse of G itself, the nested hierarchy begins

$$G_+^{-1}(y)=U+1-\frac{\alpha}{\lambda}+\frac{b_1(\lambda)}{U}+\cdots. \tag{118}$$

For every fixed N, $U/L_1^N\to\infty$; hence the shift $+1$ is beyond all orders of the outer iterated-logarithmic scale. The displacement $-\alpha/\lambda$ is smaller than the shift but larger than every inverse power of U. This four-level ordering is worth displaying explicitly:

$$\frac{U}{L_1^N}\gg 1\gg\lambda^{-1}\gg U^{-1} \qquad (N\text{ fixed}). \tag{119}$$

10.6 Numerical verification

Choose an exact z, form $y=G(z+1)$, reconstruct U,λ, and let $E_\star$ be core approximation minus exact value. Let E_j use the same sign convention after including $b_0, b_1/U, \dots, b_j/U^j$.

The core error is much larger than in the Gamma example because the linear term αz produces the slowly decaying displacement $-\alpha/\lambda$. Once that block is included, the triangular expansion again improves rapidly. At $z=5$, adding still more terms eventually encounters the asymptotic nature of the forward Bernoulli sector; at larger arguments the useful range grows.

Table 2: Signed errors (approximation minus exact value) for $\mathcal{B}(y)$, where $y = G(z+1)$.

z	$E_\star$	E_0	E_1	E_2	E_3	E_4
5	1.1187	-1.3867×10^{-2}	-7.8343×10^{-3}	-1.5935×10^{-3}	-1.8345×10^{-4}	5.8515×10^{-6}
10	6.5273×10^{-1}	-2.0081×10^{-2}	-8.4026×10^{-4}	1.5249×10^{-5}	3.2299×10^{-6}	4.1936×10^{-8}
20	4.4669×10^{-1}	-8.7205×10^{-3}	-6.6821×10^{-5}	2.3736×10^{-6}	3.4133×10^{-8}	-1.5564×10^{-9}
100	2.5342×10^{-1}	-1.2959×10^{-3}	-3.9481×10^{-7}	8.1326×10^{-9}	-5.3660×10^{-12}	-8.9993×10^{-14}

11 General phase families of Gamma–Barnes type

11.1 Polynomial-logarithmic perturbations below the leading power

A broad class covered directly by Theorem 3.1 is

$$\Phi(x) = ax^p(\log x - \beta) + C + \sum_{\rho\in A} x^{p-\rho} P_\rho(\log x) + \text{smaller sectors}, \tag{120}$$

where $A \subset \mathbb{R}_{>0}$ is well ordered and each P_ρ is a polynomial or an asymptotic coefficient transseries. Substitution $x = X(1+E)$, with $q = X^{-1}$, gives

$$\mathcal{R}(q, v, E) = \frac{1}{a}\sum_{\rho\in A} q^\rho (1+E)^{p-\rho} P_\rho\left(\beta + \frac{v}{p} + \log(1+E)\right)$$
$$+ \text{smaller normalized sectors.} \tag{121}$$

Every term has positive q-valuation. Powers of v in its coefficient are harmless because $q^\rho v^M \to 0$ for every fixed M. The inverse support lies in the additive semigroup generated by A.

This immediately explains the two examples:

Function	shifted variable	p	base support	first perturbation
$\Gamma(t+1/2)$	$t = x - 1/2$	1	$2\mathbb{N}$	$-1/(24t)$
$G(z+1)$	$z = x - 1$	2	$\mathbb{N}$	αz

For Gamma, a drop of two units in normalized power occurs because the forward term t^{-1} is divided by the order-t core. For Barnes, the forward linear term αz is divided by the order-z^2 core and produces q^1, so the full integer grid is generated.

11.2 What must be absorbed into the core

Terms with the same algebraic power x^p as the leading phase are not q-small. For example,

$$ax^p \log x + cx^p = ax^p(\log x - \beta), \qquad \beta = -c/a,$$

so the cx^p term should be absorbed exactly into β. More generally, a phase

$$x^p \left(a_r(\log x)^r + a_{r-1}(\log x)^{r-1} + \cdots\right) \tag{122}$$

requires an outer inversion in the logarithmic coefficient scale before the q-grid theorem is applied. Treating a same-power term as a small perturbation makes $\mathcal{R}(q, v, 0)$ have valuation zero and violates the hypothesis of Theorem 3.1.

This gives a practical normalization rule:

1. collect and invert the entire dominant algebraic power, including the necessary logarithmic polynomial;
2. use drops in algebraic power to define the exponentially finer q-grid;
3. place optimally truncated or resurgent exponentials in still deeper sectors.

11.3 Exact Lambert inversion for a logarithmic-power core

The exact core need not be linear in $\log x$. Let

$$\Phi_{0,r}(x) = ax^p(\log x - \beta)^r, \qquad p \neq 0, \quad r \neq 0. \tag{123}$$

Choose compatible branches of powers and logarithms. For $\Phi_{0,r}(X) = R$, define

$$Z = \frac{p}{r}\left(\frac{R}{a\mathrm{e}^{p\beta}}\right)^{1/r}, \qquad w = \mathrm{W}_k(Z), \qquad u = \frac{r}{p}w, \qquad X = \mathrm{e}^{\beta+u}. \tag{124}$$

Then $\Phi_{0,r}(X) = R$. Indeed,

$$u^r\mathrm{e}^{pu} = \left(\frac{r}{p}\right)^r w^r\mathrm{e}^{rw} = \frac{R}{a\mathrm{e}^{p\beta}}.$$

The derivative is

$$\Phi'_{0,r}(X) = aX^{p-1}u^{r-1}(pu + r). \tag{125}$$

With $x = X(1 + E)$, the normalized universal kernel becomes

$$\mathcal{K}_{p,r,u}(E) = (1 + E)^p\left(1 + \frac{\log(1 + E)}{u}\right)^r - 1, \tag{126}$$

whose linear coefficient is

$$p + \frac{r}{u} = \frac{pu + r}{u}. \tag{127}$$

The same Hahn implicit-function proof applies whenever this coefficient is a unit and the normalized perturbation has positive valuation. The case $r = 1$ reduces, after multiplication by v/p, to the kernel used in the rest of the article.

This extension covers inverse phases led by $x^p(\log x)^r$ without introducing a new special inverse function. It also clarifies the boundary of the method: a genuinely non-polynomial function of $\log x$ may still be invertible in a transseries coefficient field, but the Lambert closed form can disappear.

11.4 Coefficient algebras and nested supports

For computational work it is useful to distinguish three coefficient choices.

Rational-function coefficients. Keeping v or λ exact gives coefficients in fields such as $\mathbb{Q}(v)$ or $\mathbb{Q}(\alpha, \lambda)$. This is the compact form used for the explicit Gamma and Barnes blocks.

Laurent coefficients at large slope. Expanding each rational coefficient in v^{-1} gives an element of $\mathbb{Q}((v^{-1}))[[q^S]]$. The order is lexicographic: compare q-exponents first, then inverse powers of v. This form exposes the leading Newton response and resonant limits.

Full outer logarithmic coefficients. Substituting the expansion of $v = \mathrm{W}(Z)$ embeds the result into a larger polynomial-logarithmic transseries in L_1^{-1}, $L_2 = \log L_1$, and q. Since q itself is an exponential of $-v/p$, the resulting monomial group has at least two logarithmic-exponential levels. Expansion in this largest algebra should be postponed until presentation or comparison requires it; it is usually inefficient for coefficient generation.

11.5 Higher Barnes multiple gamma functions

The logarithms of higher Barnes multiple gamma functions have leading phases of the schematic form

$$a_m x^m \log x + b_m x^m + \text{lower algebraic powers and logarithms.} \tag{128}$$

After absorbing $b_m x^m$ into a shift of the logarithm, the core has $p = m$ and $r = 1$. Therefore the same Lambert-core and Hahn-grid machinery applies, subject to the sign and convention used for the particular multiple gamma normalization. The support semigroup is read directly from the drops in algebraic degree in the known forward expansion. Barnes G is the first nontrivial member, with $m = 2$.

12 Coefficient computation and numerical use

12.1 A generic truncated-series algorithm

The recurrence Equation (28) is readily implemented in any computer algebra system supporting truncated Hahn or Puiseux series. For an integral grid, the following pseudocode is sufficient.

Algorithm 12.1 (Triangular inversion on a q-grid)**.** Given a truncation order N, a kernel $K(E) = (v+1)E + K_{\geq 2}(E)$, and a normalized perturbation $R(q, E)$:

```
E := 0
for n := 1,...,N:
    H := truncate_q(K_ge_2(E) + R(q,E), order = n)
    e_n := - coefficient(H, q^n) / (v+1)
    E := E + e_n q^n
return E
```

At step n, every operation may be truncated immediately above q^n.

If the base support is $2\mathbb{N}$, as for shifted Gamma, replace q^n by q^{2n}. For a finitely generated fractional support, store exponents as exact rationals and process them in increasing order.

12.2 Specialized Gamma algorithm

Algorithm 12.2 (Gamma coefficient blocks)**.** Let a_k be given by Equation (54). The following procedure computes all blocks through d_{2N-1}:

```
E := 0
for n := 1,...,N:
    H := (1+E) log(1+E) - E
    for k := 1,...,n:
        H := H + a_k q^(2k) (1+E)^(-(2k-1))
    e_n := - coefficient(truncate_q(H,2n), q^(2n)) / Lambda
    E := E + e_n q^(2n)
return d_(2n-1) := e_n
```

All arithmetic is exact over $\mathbb{Q}(\Lambda)$. Bernoulli numbers are needed only up to B_{2N}.

The recurrence is output-sensitive. With naive dense series multiplication, computing N blocks takes polynomial time in N; FFT-based multiplication can reduce the high-order cost. Expression swell is controlled by retaining factored powers of Λ and combining rational functions only at the end of each step.

12.3 Specialized Barnes algorithm

Algorithm 12.3 (Barnes coefficient blocks)**.** Let $b = -1/12$ and c_k be given by Equation (86). To compute $b_0, \ldots, b_N$:

```
E := 0
for n := 1,...,N+1:
    H := K_lambda(E) - 2 lambda E
    H := H + 2 alpha q (1+E)
    H := H + 2 b q^2 (lambda+1+log(1+E))
    for k with 2k+2 <= n:
        H := H + 2 c_k q^(2k+2) (1+E)^(-2k)
    e_n := - coefficient(truncate_q(H,n),q^n) / (2 lambda)
    E := E + e_n q^n
return b_(n-1) := e_n
```

The parity check $e_n(-\alpha) = (-1)^n e_n(\alpha)$ and the limiting check Equation (113) are inexpensive regression tests.

12.4 Stable numerical evaluation

For very large y, never form $\Gamma(x)$ or $G(x)$ merely to evaluate the inverse approximation. Work with $Y = \log y$ from the beginning. The recipes are:

Gamma. Compute

$$R = Y - \tfrac{1}{2}\log(2\pi), \quad w = \mathrm{W}_0(R/\mathrm{e}), \quad T = R/w, \quad \Lambda = w + 1,$$

then evaluate Equation (61). The identity $T = \mathrm{e}^{w+1}$ is useful if R/w suffers loss of relative accuracy in a specialized numeric regime, although the quotient is normally stable on the positive real axis.

Barnes. Compute

$$R = Y - \zeta'(-1), \quad v = \mathrm{W}_0(4R/\mathrm{e}^3), \quad U = 2\sqrt{R/v}, \quad \lambda = (v+1)/2,$$

then evaluate Equation (99) and add one to recover $G_+^{-1}(y)$.

A single Newton correction in the logarithmic phase often supplies most or all of the remaining working precision; a second correction retains quadratic convergence when required. For Gamma the stable update is

$$x_{\text{new}} = x - \frac{\log\Gamma(x) - Y}{\psi(x)}, \tag{129}$$

where ψ is the digamma function. This avoids overflow from evaluating $\Gamma(x)$ itself. For Barnes G, use the analogous update with the logarithmic derivative of G, or a high-accuracy numerical derivative of $\log G$ when no dedicated routine is available.

12.5 Truncation policy

At fixed moderate argument, adding terms indefinitely is counterproductive. A robust policy is:

1. generate blocks in increasing q-order;
2. estimate the magnitude of the next block using the exact coefficient function, not only its leading large-v term;
3. stop near the smallest block, or sooner if the desired precision has been reached;

4. certify by evaluating the forward logarithmic residual and applying Equation (43) or one Newton step.

Because the Lambert core is retained exactly, far fewer terms are usually needed than in a fully expanded iterated-logarithmic formula.

13 Formal, asymptotic, and convergent statements

The calculations in this article use three notions of expansion. Their logical status should remain explicit.

13.1 Formal Hahn identities

Equations such as Equations (59) and (95) define formal solutions coefficient by coefficient. Their uniqueness is algebraic and does not depend on an analytic function realizing the forward series. Coefficient formulas, support restrictions, parity, and denominator bounds live at this level.

13.2 Poincaré asymptotic realization

The forward Gamma and Barnes expansions are Poincaré asymptotic in sectors. For every fixed truncation order, substitution of the formal inverse truncation leaves a controlled residual. Eventual monotonicity on the positive real axis and Theorem 6.1 then imply the inverse remainder. This is the meaning of Equations (61) and (99).

13.3 Convergent local inverses

The small positive Gamma branch in Equation (79) is an ordinary local analytic inverse and therefore has a nonzero radius of convergence in $s = 1/y$. The large-branch Bernoulli expansions are not of this kind. The exact special-function inverse exists analytically on its branch, but its displayed algebraic transseries is asymptotic rather than a convergent sum of all Bernoulli blocks.

13.4 Hyperasymptotic completion

The late Bernoulli coefficients encode exponentially small remainders and Stokes transitions. Forward exponentially improved expansions for Gamma and Barnes G have been developed by Nemes and earlier authors [9, 10]. Applying Equation (48) produces the leading inverse sectors. A complete resurgent treatment would additionally track alien derivatives or terminant multipliers through composition. Nothing in the algebraic grid prevents this extension, but its natural coefficient algebra is larger than the one required for the explicit results here.

14 Conclusions

The large inverse branches of Gamma and Barnes G have a common normal form. Taking logarithms exposes a superlinear phase; a Lambert W expression inverts its $x^p \log x$ core exactly; a relative displacement then satisfies an implicit equation with a unit linear coefficient. The remaining inversion is triangular in a Hahn grid.

For Gamma, the half-shift $x = t + 1/2$ removes every even inverse power from the forward Stirling phase. The inverse displacement consequently has only odd powers $T^{-1}, T^{-3}, \ldots$, with coefficient polynomials in Λ^{-1}. The recurrence Equation (62) computes them to arbitrary order and the forward remainder transfers with one additional factor $1/\Lambda$.

For Barnes G, the linear term creates the first displacement $-\alpha/\lambda$, while the logarithmic term is resonant. Its surviving large-slope limit sums to the elementary algebraic series $\sqrt{1 + q^2/6} - 1$. This both explains the coefficient pattern and provides a strong symbolic check. The actual inverse G_+^{-1} is obtained by restoring the shift $+1$.

The general calculus applies whenever the dominant phase can be normalized to $ax^p(\log x - \beta)$, or more generally to $ax^p(\log x - \beta)^r$, and all unabsorbed algebraic corrections have positive relative power drop. Its essential ingredients are exact core inversion, a well-ordered support semigroup, a unit normalized slope, and a residual certificate. These conditions are constructive: they yield coefficient algorithms rather than merely existence statements.

Summary

For practical use, keep Lambert W unexpanded, center the forward phase so that its support is as sparse as possible, compute the inverse blocks by the triangular recurrence, and validate the chosen truncation by its logarithmic residual. Expand into iterated logarithms only as a final presentation layer.

A Coefficient extraction identities

This appendix records two identities useful for independent implementations.

A.1 Kernel coefficients

From Equation (24),

$$\kappa_n(p, v) = \frac{v}{p}\binom{p}{n} + \binom{p}{n}\sum_{j=0}^{n-1}\frac{1}{p-j}, \tag{130}$$

whenever none of $p, p-1, \ldots, p-n+1$ is zero. The right-hand side extends polynomially through removable singularities because it is simply $\frac{v}{p}\binom{p}{n} + \partial_p\binom{p}{n}$. For integral p, it is safer to evaluate the derivative form before substitution.

For Gamma, $p = 1$ and the universal nonlinear part simplifies to

$$(1+E)\log(1+E) - E = \sum_{m\geq 2}\frac{(-1)^m}{m(m-1)}E^m. \tag{131}$$

For Barnes, direct expansion gives

$$\mathcal{K}_\lambda(E) = 2\lambda E + \lambda E^2 + E^2 + \frac{1}{3}E^3 - \frac{1}{12}E^4 + \frac{1}{30}E^5 - \cdots. \tag{132}$$

The separation of $2\lambda E$ in the recurrence is essential; the term λE^2 remains nonlinear and is responsible for the algebraic resonant limit.

A.2 Bell-polynomial form of the small Gamma branch

Let

$$\log\Gamma(1+x) = \sum_{m\geq 1}\ell_m\frac{x^m}{m!}, \qquad \ell_1 = -\gamma, \qquad \ell_m = (-1)^m(m-1)!\zeta(m) \quad (m\geq 2). \tag{133}$$

If Y_n denotes the complete exponential Bell polynomial, then

$$\frac{\mathrm{d}^{n-1}}{\mathrm{d}x^{n-1}}\Gamma(1+x)^n\bigg|_{x=0} = Y_{n-1}(n\ell_1, n\ell_2, \ldots, n\ell_{n-1}). \tag{134}$$

Consequently,

$$\Gamma_0^{-1}(y) = \sum_{n\geq 1} \frac{Y_{n-1}(n\ell_1, \ldots, n\ell_{n-1})}{n!\, y^n}. \tag{135}$$

This gives exact symbolic coefficients to arbitrary order without performing series composition explicitly.

The next coefficient after Equation (81) is

$$[y^{-6}]\,\Gamma_0^{-1}(y) = -\biggl(\frac{54\gamma^5}{5} + 3\gamma^3\pi^2 + 6\gamma^2\zeta(3) + \frac{17\gamma\pi^4}{120} + \frac{\pi^2\zeta(3)}{6} + \frac{\zeta(5)}{5}\biggr). \tag{136}$$

B The sixth displayed Barnes block

For completeness, the coefficient multiplying U^{-5} in Equation (99) is

$$b_5(\lambda) = \frac{P_5(\lambda, \alpha)}{362880\lambda^{11}}, \tag{137}$$

where

$$\begin{aligned} P_5 = {} & 22680\alpha^6\lambda^5 + 40824\alpha^6\lambda^4 - 349272\alpha^6\lambda^3 - 352800\alpha^6\lambda^2 \\ & + 1111320\alpha^6\lambda - 476280\alpha^6 + 11340\alpha^4\lambda^7 + 34020\alpha^4\lambda^6 + 3780\alpha^4\lambda^5 \\ & - 138600\alpha^4\lambda^4 - 132300\alpha^4\lambda^3 + 132300\alpha^4\lambda^2 + 1890\alpha^2\lambda^9 \\ & + 8568\alpha^2\lambda^8 + 15120\alpha^2\lambda^7 + 11718\alpha^2\lambda^6 - 3150\alpha^2\lambda^5 - 9450\alpha^2\lambda^4 \\ & + 105\lambda^{11} - 668\lambda^{10} - 378\lambda^9 - 21\lambda^8 + 175\lambda^7 + 105\lambda^6. \end{aligned} \tag{138}$$

The coefficient satisfies the predicted even parity in α, and

$$\lim_{\lambda\to\infty} b_5(\lambda) = \frac{105}{362880} = \frac{1}{3456}, \tag{139}$$

matching Equation (114).

C Direct residual checks

The coefficient formulas can be checked without evaluating either inverse function. For Gamma, substitute

$$E_N(q) = \sum_{n=1}^{N} d_{2n-1}(\Lambda) q^{2n} \tag{140}$$

into the left-hand side of Equation (59). The defining recurrence is equivalent to

$$\Lambda E_N + (1 + E_N)\log(1 + E_N) - E_N + \sum_{k=1}^{N} a_k q^{2k}(1 + E_N)^{-(2k-1)} = O(q^{2N+2}). \tag{141}$$

The coefficient of q^{2N+2} predicts the next block before any special function is evaluated.

For Barnes, put

$$E_N(q) = \sum_{n=1}^{N+1} b_{n-1}(\lambda) q^n. \tag{142}$$

Then the left-hand side of Equation (95), with the Bernoulli sum truncated wherever its exponent does not exceed $N+1$, is

$$O(q^{N+2}). \tag{143}$$

These identities are exact in the rational-function coefficient field and serve as stronger symbolic tests than decimal comparison.

For analytic verification at a numeric target, evaluate instead

$$\delta_\Gamma = \log\Gamma(\widehat{x}) - \log y, \qquad \delta_G = \log G(\widehat{x}) - \log y, \tag{144}$$

and divide by the corresponding logarithmic derivative. This avoids the severe scaling and overflow that would result from subtracting the original function values.

D Notation dictionary

Symbol	Meaning
Y	Logarithmic target, usually $Y = \log y$.
C	Constant term removed from the forward phase.
$R = Y - C$	Target for the exact dominant core.
p, a, β	Parameters of $ax^p(\log x - \beta)$.
v	Lambert variable $\mathrm{W}((pR/a)\mathrm{e}^{-p\beta})$.
X	Exact Lambert core $\mathrm{e}^{\beta}\mathrm{e}^{v/p}$.
$q = X^{-1}$	Exponentially small grid monomial relative to v^{-1}.
ϵ	Relative inverse displacement, $x = X(1+\epsilon)$.
T, Λ	Gamma core and slope parameter: $T = R/\,\mathrm{W}(R/\mathrm{e})$, $\Lambda = \log T$.
U, λ	Barnes core and half-slope parameter: $U = 2\sqrt{R/v}$, $\lambda = (v+1)/2$.
d_{2n-1}	Gamma displacement coefficient multiplying $T^{-(2n-1)}$.
b_n	Barnes displacement coefficient multiplying U^{-n}.
C_G	Barnes constant $\zeta'(-1) = 1/12 - \log A$.

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